

Chapter 19 — Energy Modalities in Urology

Chapter Notes · Bailey, Forbes, Milner and Teichman

- Energy modalities deliver **five surgical effects**: cutting, tissue removal (ablation), tissue destruction, coagulation and fragmentation.
- **Safe outcomes require informed use of each modality**. The chapter's emphasis is the **tissue-energy interaction mechanism and the safety** of each.
- Organising perspective: **the universality of the wave nature of physical processes** explains most energy-based devices in current use.

IN PLAIN TERMS — THE PHYSICS YOU NEED BEFORE STARTING

Boxes titled "In plain terms" are explanatory additions for understanding. They are not Campbell-Walsh-Wein text; examinable content is in the notes around them.

- **Energy** is the capacity to do work: heat tissue, crack a stone, move something. It is measured in **joules (J)**. One joule is roughly the energy needed to lift a 100-g object by one metre.
- **Power** is **how fast energy is delivered**, measured in **watts (W) = joules per second**. A 100-W laser delivers 100 J every second. The same energy delivered faster means more power. On a lithotripsy laser, **power = pulse energy × pulses per second**: $0.5 \text{ J} \times 20 \text{ Hz} = 10 \text{ W}$.
- **Intensity** is **power per unit area**. Sunlight spread over your hand is harmless; the same sunlight concentrated by a magnifying glass burns paper. The power is the same, but the intensity is far greater.
- **Pressure** is force per unit area, measured in pascals or bars. **1 bar is about atmospheric pressure**, so a lithotripter shock of 1000 bar is about a thousand atmospheres, applied for a millionth of a second.
- A **wave** is a disturbance that **carries energy from place to place without carrying the material with it**. In a stadium "Mexican wave", each spectator only stands and sits in place, yet the wave travels round the stadium.
- **Sound (acoustic, elastic) waves**: the material itself is alternately **squeezed and stretched**, with each molecule pushing its neighbour. Sound therefore **needs a material** (water, tissue, stone) and cannot cross a vacuum. **Ultrasound is simply sound too high-pitched to hear**. Lithotripter shocks, ultrasound imaging, HIFU and histotripsy are all sound.
- **What "electromagnetic" means**: an **electric field** is what makes charged objects push or pull on each other (static cling). A **magnetic field** is what turns a compass needle. A changing electric field creates a magnetic field, and a changing magnetic field creates an electric field, so the two keep regenerating each other and **travel as a wave, even through empty space**. That is why sunlight reaches us across a vacuum.
- **Radio waves, microwaves, infrared (the heat felt from a fire), visible light, ultraviolet and X-rays are all the same phenomenon**, differing only in wavelength. Lasers (light), microwave ablation and radiofrequency energy are therefore **cousins**, and the X-rays of your fluoroscopy are the same family.
- **Frequency** is how many waves pass per second (hertz, Hz). **Wavelength** is the length of one wave. For a given speed, **higher frequency means shorter wavelength** (speed = frequency × wavelength).
- You already use this trade-off daily: a **high-frequency linear probe** (scrotum) shows fine detail but only superficially, and a **low-frequency curvilinear probe** (kidney) reaches deep but with less detail. **High frequency = shallow; low frequency = deep** runs through the whole chapter.
- **Absorption** is a wave giving up its energy to the material it passes through, mostly as heat. **Where energy is absorbed is where the surgical effect happens**. Water absorbs holmium laser light strongly,

which is why Ho:YAG acts only millimetres from the fibre tip. Haemoglobin absorbs green light, which is why GreenLight works on the vascular prostate.

- A **pulse** is a short burst of energy rather than a continuous stream; **pulse duration** is how long each burst lasts. **The same energy squeezed into a shorter burst gives a far higher peak power**, like a hammer blow compared with a steady push.
- **Size prefixes**: milli (ms) = one-thousandth; micro (μ s) = one-millionth; nano (ns) = one-billionth; femto (fs) = a million-billionth. kilo (k) = thousand; mega (M) = million; giga (G) = billion. A 250- μ s pulse lasts a quarter of a millisecond.

WAVES

Two modes of energy delivery

- **Directed energetic waves**, or **direct contact** (water vapour, cryosurgery).

ELECTROMAGNETIC VS ELASTIC WAVES

	Electromagnetic (EM) waves	Elastic waves
Devices	Lasers, electrocautery, microwave, radiofrequency	Shock wave lithotripsy, ultrasound therapies, intracorporeal lithotripters
Nature	Oscillation between electric and magnetic fields	Mechanical vibration : oscillation between pressure (potential energy) and particle velocity (kinetic energy)
Media	Free space and media	Gases, liquids and solids (including stones)
Subclasses	By wavelength band	Acoustic vs ultrasonic by frequency; longitudinal (motion along propagation) vs shear (motion transverse to it)
Wavelength	Micron order (light)	Millimetre order (medical acoustics)

IN PLAIN TERMS — READING THE WAVE DESCRIPTORS

- Picture a wave on a rope you are shaking. **Amplitude** is the height of the wiggle (how strong the wave is). **Frequency** is how many times per second you shake. **Wavelength** is the distance between two crests.
- **Phase** is where a point sits in the cycle: crest, trough or halfway. Two waves "in phase" rise and fall together and reinforce each other; out of phase, they cancel.
- **Polarisation** is the direction of shaking, up-down or side-to-side.
- **Longitudinal vs shear**: pushing a Slinky along its length sends a compression along it (longitudinal: sound in water and tissue). Flicking a rope sideways sends a sideways ripple (shear). **Liquids cannot carry shear waves; solids can**. This is why shear waves inside a stone matter in lithotripsy.
- **Dispersion** means different frequencies travel at different speeds in a material, so a mixed pulse spreads out, like runners who start together but finish strung out. A glass prism splitting white light into a rainbow is dispersion.

WAVE DESCRIPTORS

Descriptor	Definition	Unit / symbol
Frequency	Inverse of the period (T), the time for one oscillation	Hertz (Hz) = s^{-1}
Wavelength	Spatial distance of one oscillation	λ
Wavevector	Change of phase over distance; $k = 2\pi/\lambda$	k, radians per unit length
Amplitude	Maximum oscillation of the wave	Pressure for acoustics

Descriptor	Definition	Unit / symbol
Phase	Dimensionless position within the period or wavelength	Degrees or radians
Polarisation	Direction of oscillation relative to propagation	—
Dispersion	Property of the medium: dependence of frequency on wavelength or wavevector	$f(k)$; in EM written $\omega(k)$

- A wave is **propagating energy oscillating between potential and kinetic forms** (ocean wave: water height $\leftrightarrow$ motion).
- Elastic pressure and velocity are both expressed in **megapascals (MPa)**.

TABLE 19.1 PARAMETERS AND UNITS

Quantity	SI unit	Relation
Force	Newton (N)	—
Pressure or stress	Pascal (Pa); bar	1 Pa = 1 N/m²; 10 bar = 1 MPa
Energy	Joule (J)	1 J = 1 N·m
Power	Watt (W)	1 W = 1 J/s
Intensity	W/cm ²	Power per unit area

- What the table is for: **pressure is the amplitude of acoustic therapy, and intensity is its delivered dose.** Lithotripsy papers quote bar, physics papers quote MPa.
- The trap: **10 bar = 1 MPa, not 1 bar = 1 MPa.**

IN PLAIN TERMS — SQUEEZE, PULL AND CAVITATION

- A sound wave **alternately squeezes (positive pressure) and pulls apart (negative pressure)** the medium it passes through.
- Liquids resist squeezing very well but can be **torn by pulling**. When the pulling phase is strong enough, the liquid rips open into a vapour bubble. This is **cavitation**, the same bubble that appears when you pull hard on a syringe plunger against a closed tip.
- **Boiling makes bubbles by heating; cavitation makes bubbles by pulling.**
- Cavitation matters throughout the chapter. It **helps break stones**, it **injures tissue**, and when too many bubbles form it **blocks the energy** from reaching the target.

Pressure, negative pressure and cavitation

- Real waves are **never a pure single frequency**; they are sums of harmonic components.
- For acoustics, **amplitude (loudness) is pressure**. It **can be negative**, because the fluid is elastic enough to hold its phase through the negative half-cycle.
- A sufficiently intense negative phase causes **cavitation**, a **liquid–gas phase change**. **Cavitation vs boiling:** in cavitation the **pressure drops**; in boiling the **temperature rises**.
- Spherical bubble dynamics in an incompressible fluid follow the **Rayleigh–Plesset equation**, derived from the **Navier–Stokes equation**.

IN PLAIN TERMS — ONE NOTE VERSUS A HAND-CLAP

- A **single sustained flute note is narrowband**: essentially one pitch. A **hand-clap is broadband**: a sharp sound containing many pitches at once. **The shorter and sharper a sound, the more frequencies it must contain.**
- The **Fourier transform is a mathematical prism**. As a prism splits white light into its colours, Fourier analysis splits any pulse into the pure frequencies it is built from.

- The **shock wave lithotripsy pulse is a clap (broadband)**; the **burst wave lithotripsy pulse is a short sung note (narrowband)**.

BROADBAND VS NARROWBAND

Broadband	Narrowband
Short pulses	Long-duration pulses or bursts of many cycles at one frequency
Contain many frequencies	Concentrated at one centre frequency

- **Truncating a continuous wave increases its bandwidth**; the original frequency becomes the **centre frequency**.
- A **pulse** is a packet of sound or light waves (a tone burst or a simple burst). **Therapy is delivered as an ensemble of pulses or bursts**.

Fourier transform (Equation 1)

$$p(t) = \int P(f) \exp(j2\pi ft) df$$

- Any pulse $p(t)$ **resolves into a sum (integral) of harmonics**, each with amplitude $P(f)$. The transform shows **which frequencies a pulse contains**.

IN PLAIN TERMS — WHAT A "SHOCK" IS

- In a very loud wave the high-pressure crests travel slightly faster than the troughs. The front of the wave therefore **steepens until it becomes an almost vertical jump in pressure**. That jump is a shock, the same thing as a sonic boom.
- A steep jump means an **enormous pressure change in a tiny time**, and that is what cracks brittle stones.
- The word "**shock**" **has nothing to do with electricity**.

Shock formation

- A **shock is a discontinuous increase in pressure**. It forms **at high amplitude, especially at high frequency over long distances**.
- Mechanism: **nonlinearity**. Wave speed depends on amplitude, so **peaks travel faster than troughs** and the peak catches the preceding part of the wave. It forms a shock but **does not "crash"** like an ocean wave.
- Frequency-domain signature: **energy transfer from the fundamental (centre) frequency into higher harmonics**.
- Medical acoustic waves lie **well above 20 kHz** (limit of human hearing), but some contain audible components. **What is heard is the click rate of pulse delivery, not the centre frequency**.
- **Duty cycle** = percentage of total time that sound is emitted.

Dispersion in EM

- **EM wave speed depends on frequency**, according to the medium's dispersion relation $\omega(k)$, where ω is the angular frequency ($2\pi f$, radians/s) and k the wavevector.
- Consequence: **a pulse sorts by frequency**. Components arrive with different delays and the **pulse shape changes as it propagates**.

IN PLAIN TERMS — FOCUSING IS THE MAGNIFYING GLASS

- A magnifying glass **concentrates the same sunlight into a tiny spot** hot enough to burn paper. Lithotripters and high-intensity focused ultrasound do exactly this with sound.

- Where the beam is wide (at the skin) the energy is spread and harmless. Where it converges (at the stone or target) the energy is concentrated and effective. **This is how energy passes through skin and fat without injuring them.**
- The focal spot is about one wavelength wide and about ten wavelengths long, giving a **cigar-shaped or rice-grain-shaped zone** of effect.

Focusing and beams

- A **point source** (smaller than the wavelength) **radiates spherically**; a lens or reflector **focuses waves into a beam**.
- Beams decompose into **spatial modes**, each of which diffracts and propagates — analogous to the temporal Fourier decomposition.
- **Beam width = full width at half maximum (FWHM)**: the distance between points where amplitude falls to half the peak; it can also be measured on intensity.
- The commonest laser beam is the **Gaussian beam**, whose amplitude varies with radius as a Gaussian function.

IN PLAIN TERMS — PHOTONS

- Light comes in tiny indivisible packets called **photons**.
- **The colour (wavelength) of a laser fixes the energy of each photon; its power setting fixes how many photons leave per second.** Turning up the power sends more photons, not stronger ones.

Intensity, power and photons

- **Intensity $\propto$ amplitude²**, so it is **always positive** even though amplitude oscillates.
- **Power = energy per unit time. Intensity = power per unit area**, the rate of energy passing through the beam cross-section; often averaged over on-time or area.
- Photon energy is related to frequency by Planck's constant (h): **$E = hf$. A more powerful laser produces more photons per unit time, not more energetic photons.**
- **Phonons** are the acoustic analogue; the term is not used in medical ultrasound.

IN PLAIN TERMS — WHY GEL, WHY SHADOWS

- **Acoustic impedance** is essentially how stiff and dense a material is to sound.
- Sound passes easily between materials of similar impedance (water to soft tissue) but **bounces back from very different ones** (tissue to air, or tissue to bone or stone).
- This explains four things you already know:
 - **First**, you apply gel, because even a thin layer of air between probe and skin reflects almost everything.
 - **Second**, bowel gas and ribs cast shadows on ultrasound.
 - **Third**, a stone appears bright with an acoustic shadow behind it.
 - **Fourth**, a lithotripter must be coupled to the patient with water or gel.
- **Refraction** is the bending of a wave as it enters a new material at an angle, the reason a straw looks bent in a glass of water.

Scattering, diffraction, refraction and reflection

- **Diffraction**: free-space spreading from an aperture, computed with the Fraunhofer and Kirchhoff relations.
- **Scattering** occurs wherever the **dispersion relation changes** — a different state (pressure, temperature), a different phase, or a different material.

- **Refraction:** at oblique incidence the ray bends on entering a layer with a different refractive index (EM) or wave speed (elastic).
- At an interface, part of the energy is **transmitted** and part **reflected**. **Acoustic impedance = sound speed × density** governs the ratio.
- **Large impedance mismatch → reflection, poor transmission.** Air (bowel gas) and bone reflect ultrasound.
- **Rough surface** (irregularities smaller than the wavelength) → scatter in many directions; a **smooth surface** gives specular reflection.

IN PLAIN TERMS — WHY ENERGY FADES WITH DEPTH, AND WHY WAVES PUSH

- Every material "eats" part of a wave as it travels, either by **absorbing** it (turning it into heat) or by **scattering** it in other directions. Together these are **attenuation**.
- The **absorption distance** is how deep the wave goes before most of its energy is gone. A **short absorption distance dumps all the energy close by**: intense local heating, but safe for deeper structures.
- **Radiation force:** a wave also gives a small push to whatever it hits, like wind on a sail. For light the push is negligible. For sound it is large enough to be useful, and **ultrasonic propulsion literally pushes stones**.

Attenuation and absorption

- **Attenuation = scattering + absorption.**
- **Absorption reduces amplitude by $e^{-\alpha z}$** , where α is the absorption coefficient (1/distance) and z the propagation distance.
- **Absorption distance = $1/\alpha$** : the depth at which amplitude or energy falls to $1/e$.
- **Absorption usually increases with frequency.** The acoustic absorption distance in tissue is **about 25 wavelengths**.
- A short absorption distance cuts both ways: it **protects deeper structures** but **concentrates heat** in a small, engineerable region.
- Absorbers: **dark objects** (at the incident wavelength) absorb EM; **viscous material** absorbs acoustic waves.

BIOEFFECT MECHANISMS: EM VS ACOUSTIC

EM waves	Elastic / acoustic waves
Mostly absorption → heat	Thermal absorption, mechanical stress (in solids), or cavitation (in fluids and tissue)

Radiation force

- Absorption also **imparts a force along the propagation direction**.
- **Radiation force = power absorbed ÷ wave speed.** It is **much smaller in EM**, because light is far faster than sound.
- It is **doubled when the object is totally reflective**, because the momentum is reversed rather than absorbed.
- When the **wavelength is much longer than the distances of interest**, modal analysis replaces ray thinking.

Figure legends

- **Fig. 19.2** — Defocusing (point source) vs focusing (lens or reflector), with beam width, cross-section, focal region, ray line and focal distance f . **Focal width ≈ 1 wavelength; axial extent ≈ 10 wavelengths.**
- **Fig. 19.3** — Holmium:yttrium-aluminium-garnet (Ho:YAG) output from a 365- μm fibre is **near-Gaussian**; **photon density correlates with ablation crater volume and geometry.** (Lee, Ryan, Teichman et al., *J Endourol* 2003.)

- **Fig. 19.4** — EM and acoustic spectra. The **laser wavelength map** runs from ultraviolet to mid-infrared: excimer, argon, potassium-titanyl-phosphate (KTP), pulsed dye, ruby, alexandrite, diode, neodymium:YAG (Nd:YAG), thulium fibre laser (TFL), thulium:YAG (Tm:YAG), Ho:YAG, erbium:YAG (Er:YAG), carbon dioxide (CO₂). Acoustic bands: infrasound <20 Hz; audible to 20 kHz; ultrasound above. **Medical therapy ≈ 2 MHz region; diagnostics up to ≈ 200 MHz (acoustic microscopy). Medical ultrasound commonly 100 kHz to 20 MHz. Sound speed depends on the medium**, so a given frequency has a longer wavelength in a stone than in water (≈3× in cystine).

KEY POINTS

- **Lithotripsy shock waves are acoustic waves with abrupt pressure rises, not electrical shocks.**
- "Frequency" means either the **oscillation frequency within a pulse** or the **pulse delivery rate**; the latter is lower.
- A pulse has a **duration in time** and a **length in space**, linked by the wave speed.
- **Short pulses are broadband; long pulses are narrowband.**
- **EM bioeffects = absorption → heat. Acoustic bioeffects = heat, mechanical stress (solids), cavitation (fluids/tissue).**

Concept — every modality in this chapter is a question of **where the wave's energy is absorbed and in what form it is released**. Frequency sets the penetration, impedance sets what is reached, and absorption sets whether the result is heat, stress or a bubble.

ELECTROMAGNETIC WAVES

Lasers

IN PLAIN TERMS — HOW A LASER WORKS

- An **ordinary bulb** is heated atoms giving off light at random: every colour, every direction, out of step, like a crowd chatting.
- A **laser** energises ("pumps") atoms and then triggers them to release light **in lockstep**: one colour, one direction, in step, like a choir singing one note in unison.
- **Mirrors at each end** bounce the light back and forth through the material to build it up, and one mirror lets a little escape as the beam.
- Result: a **pure single wavelength** in a **tight beam** that can be sent down a thin fibre with enormous concentration.
- **The laser's name is the material that makes the light**: Ho:YAG = holmium atoms in a YAG crystal; Tm = thulium; KTP/LBO = crystals that halve the wavelength to make green light.
- **The wavelength decides what absorbs the light** (water, haemoglobin, pigment). That is the whole reason different lasers do different things.

Principle

- **LASER = light amplification by stimulated emission of radiation**. A **gain medium** (gas, solid or liquid) sits inside a **resonant optical cavity**.
- The medium is "**pumped**" by an external source: electrical current, chemical energy, a flashlamp or another laser.
- **Stimulated emission**: pumped electrons in uniform higher-energy orbitals → an incoming photon stimulates release of a photon with **identical energy, propagation direction and phase** → cascade → output.

LASER VS ORDINARY LIGHT

Laser (coherent)	Blackbody / incandescent (incoherent)
Narrow wavelength range , narrow angular spread, smooth wavefronts	Broad continuum of wavelengths, wide angular spread
Radiance (W/cm²·sr) orders of magnitude greater	Household bulb, the sun

Resonator

- Two **inward-facing mirrors** act as feedback elements; photons pass repeatedly through the gain medium, stimulating further emission, **until amplitude saturates**.
- The opposed mirrors give **directional, in-phase output**: a **collimated, monochromatic, coherent beam**.
- In practice: **one total reflector (ideally 100%)** and **one partial reflector** that lets a small fraction escape — the **output end**.

IN PLAIN TERMS — TAPS AND DAMS

- **Continuous wave** is a tap left running.
- **Q-switching** is a dam: the energy is held back while it builds up and then released all at once as one huge, very brief flood.
- **Mode-locking** produces even shorter flashes.

LASER OUTPUT MODES

Mode	Mechanism	Pulse duration
Continuous wave (CW)	Steady leak through the partial reflector	Continuous
Q-switched	Shutter (Venetian-blind or electro-optic) closed while the gain medium saturates, then opened → giant pulse	Nanoseconds
Mode-locked (phase-locked)	Constructive/destructive interference in the cavity	Few to tens of femtoseconds

Diode lasers

- **Semiconductor gain medium** with mirrors fabricated on the semiconductor; a doped wafer (holes and electrons) creates a charge bias, and the diode is **pumped by electrical current**.
- Advantages: **smaller, more energy-efficient**, and **configurable across a wide range of wavelengths and pulse durations**.
- Historically non-medical (barcode scanners, CD/DVD players). Clinical diodes run at **800–980 nm**, mainly for **soft tissue**.

History

- **First lithotripsy laser: ruby crystal with flashlamp pump (1960)**. Medical lasers must be **mobile, serviceable and robust to clinical use**.
- **CO₂ laser (1964)**: gas laser, **10,600 nm**; **very highly absorbed by water** → **narrow penetration**. Uses: penile and vulvar lesions (superficial squamous cell carcinoma), skin resurfacing.
- **Not usable for intracorporeal lithotripsy**: 10.6-μm light suffers fibre losses, and hollow waveguides fail in aqueous environments.

Figure legend

- **Fig. 19.5** — (A) Lasing medium energised by a flashlamp; light oscillates between mirrors, gaining energy each pass; the partly transparent mirror 2 emits a collimated beam. (B) Coherent radiation: smooth, usually

spherical wavefronts, narrow angular and frequency distribution. (C) Incoherent blackbody radiation: wide angular and frequency distribution. (Teichman, *Smith's Textbook of Endourology*, 2007.)

Laser Lithotripsy

IN PLAIN TERMS — THE TWO WAYS A LASER BREAKS A STONE

- **Photomechanical ("hammer")**: an extremely short pulse (billionths to millionths of a second) dumps energy so fast it cannot escape as heat. Instead it **explodes, creating a bubble and a shock wave that crack the stone mechanically**. Pulsed dye and FREDDY lasers worked this way.
- **Photothermal ("blowtorch" or "drill")**: a longer pulse (hundreds of microseconds: holmium and thulium) **heats the spot it hits** until the stone melts, decomposes, or its internal water flashes to steam. The stone is **eroded into dust and small fragments**.
- **Pulse duration decides which happens, and modern holmium and thulium lithotripsy is photothermal**.
- **Why the "pulse duration" on the machine can mislead**: manufacturers quote how long the electrical current pumping the laser lasts, not how long the light actually lasts at the stone.

Pulse duration — the key parameter

- Lithotripsy pulses span **femtoseconds to microseconds**.
- **Nomenclature trap**: manufacturers usually quote the **electrical pulse duration** of the pump current; the **optical pulse duration (FWHM)** is the physically informative value; labels such as "**short / medium / long**" **differ between platforms**.

First principle

- **Fragmentation begins with photon absorption**. No absorption in the stone or surrounding water → **no fragmentation**.
- The **most important factor is fluence**: optical energy absorbed by the stone and/or water. Ablation is multifactorial: **photothermal plus photoacoustic/photomechanical**.

LASER PULSE-DURATION CLASSES AND MECHANISMS

Class (optical pulse)	Dominant mechanism	Bubble	Pressure transient	Fragments
Ultrashort <500 ns (Q-switched)	Plasma (needs irradiance >10 ⁸ W/cm ²)	Spherical	Extremely high at fibre tip	Larger than cavitation (femtosecond, research only: fine debris)
Short >500 ns to <10 µs	Cavitation and shock waves (photomechanical)	Spherical, single-locus collapse	High (>300 bar)	Large
Transition ≈10–100 µs (photoacoustic predominates to ~50 µs)	Mixed photoacoustic + photothermal	Intermediate	Intermediate	—
Long >250 µs (Ho:YAG, TFL, pulsed Tm:YAG)	Photothermal	Asymmetric, multi-locus collapse	Weak (<20 bar)	Small
Very long >20 ms / CW	Photothermal with collateral coagulation and carbonisation	—	—	Adjacent tissue injured

IN PLAIN TERMS — BUBBLES, PLASMA AND "STRESS CONFINEMENT"

- **Cavitation bubble**: the laser tip boils a pocket of water into a vapour bubble that balloons outward and then **collapses violently (implodes)**, sending out a pressure wave.

- The punch of the collapse rises with the **cube of the bubble radius**: double the bubble size gives eight times the punch.
- **Plasma**: at extreme intensity the material is torn into charged particles, a tiny spark like miniature lightning, whose sudden expansion hits the stone.
- **Spallation**: a pressure wave travelling through a stone reflects from its far surface as a pulling wave and flakes material off that surface, like a chip blown off the far side of a wall that is struck.
- **Stress confinement** means delivering the blow **faster than the stone can relieve it**. A karate chop breaks a board that a slow push would only bend.

Photoacoustic / photomechanical mechanisms (pulses <50 μs)

- Sudden thermoelastic expansion, optical breakdown or plasma initiation convert laser energy into **stress waves (at sound speed) or shock waves (faster)**.
- **Water-absorbed light** → spherical cavitation bubble → single-locus collapse → outgoing shock wave.
Stone-absorbed light → thermoelastic expansion within the stone → internal fracture.
- **Three generators of transient stress waves**:
 1. **Thermoelastic expansion** → pressure wave reflects at the free surface as an inward **negative (tensile) wave** → if strong enough, **spallation** (disruption from rapid heating or cooling; rapid cooling gives **irregular fragmentation**).
 2. **Recoil**: ejected ablated material → conservation of momentum → recoil pressure wave → **stress fracture deep to the ablation surface**.
 3. **Phase change**, e.g. **plasma**, the "fourth state of matter" (an ionised cloud of free electrons); unstable rapid expansion and contraction → **stress fractures in the stone matrix**.
- Laser shock waves arise **twice per pulse**: at the **outset** (bubble expansion or plasma formation) and at the **end** (spherical bubble collapse).

Cavitation bubble dynamics (ultrashort and short pulses, up to ~100 μs)

- Laser-induced water vapour expands spherically at the fibre tip, then **collapses in the incompressible surrounding water hundreds of microseconds later**.
- Collapse releases the stored potential energy instantaneously → strong pressure waves → **strain on the crystal matrix** → **fragmentation**.
- **Pressure-transient sequence**: initial transient (bubble expansion) → end of laser pulse → bubble collapse → **second, higher transient** → fragmentation.
- **Transient amplitude** $\propto$ (**maximum bubble radius**)³.

Stress confinement

- Efficient photomechanical fragmentation requires **energy delivered within a pulse shorter than the time for stress to propagate out of the irradiated zone** → rapid pressure and heat build-up, often a shock wave, and **dissociation of the crystalline structure**.

PHOTOMECHANICAL LASERS: PULSED DYE VS FREDDY

	Pulsed dye (coumarin)	FREDDY
Full name	Organic-dye laser, flashlamp-pumped	Frequency-doubled double-pulse Nd:YAG
Gain medium	Organic dye (liquid); tunable	Solid-state : KTP crystal inside an Nd:YAG resonator
Wavelength	504 nm	1064 nm + 532 nm simultaneously
Pulse	<2 μs	1.2 μs

	Pulsed dye (coumarin)	FREDDY
Safety	Haemoglobin absorbs 504 nm → stray energy carried off by blood flow ("energy sink") → high safety margin	—
Limitation	Large fragments; poor on hard stones (cystine, brushite, calcium oxalate monohydrate)	—

- **Solid-state lasers** use a **crystalline host doped with ions** as the gain medium.

Figure legend

- **Fig. 19.6** (Jansen et al., *Lasers Surg Med* 1996) — Ho:YAG ($\lambda = 2.1 \mu\text{m}$) bubble expansion and collapse in water, fast-flash photography, at **500 ns, 100 μs , 260 μs , 460 μs and 1100 μs** . The 500-ns pulse used a **200- μm fibre at $14 \pm 1 \text{ mJ}$** ; all others a **400- μm fibre at 200 mJ**. Right column: maximum bubble size in **84% water polyacrylamide gel — always smaller than in water**. (A) 500 ns: expansion wave at 1 μs , collapse wave at 215 μs . (B) 100 μs : collapse wave at 300 μs . (C) 260 μs (210 μs FWHM): collapse wave at 270 μs . (D) 460 μs (410 μs FWHM). (E) 1100 μs (1050 μs FWHM): **oscillating bubble between 400 and 1000 μs** .

Long Pulse Durations ($\geq 250 \mu\text{s}$) and Photothermal Ablation

IN PLAIN TERMS — THE MOSES EFFECT, FIBRE ANGLE AND WHY WATER MATTERS

- Water absorbs holmium and thulium light strongly, so a fibre held a few millimetres from the stone would waste its energy heating water.
- What actually happens: the **first part of each pulse boils a tunnel of steam** between fibre and stone. Steam absorbs this light far less than liquid water, so **the rest of the pulse passes through the tunnel to the stone**, as the sea parted for Moses.
- Practical consequences: **keep the fibre close to the stone**, and **fire straight at it (perpendicular)**, not **grazing its surface**. A photothermal laser works only if its photons actually enter the stone.
- **Why long pulses give so little pressure**: the bubble forms slowly and lopsidedly and collapses in several places at different times, so there is no single violent implosion.
- **Mid-infrared (thulium and holmium) light is highly absorbed by water**. Pulses of **several hundred μs → photothermal removal** by rapid vaporisation, melting, carbonisation or chemical decomposition, **confined to the zone of absorption**.
- Pulses **>20 ms, or CW** → heat conducted to surrounding tissue → **coagulation and carbonisation**; lithotripsy pulses are therefore kept **well below 20 ms**.
- **Moses effect: mid-infrared light is absorbed more by liquid water than by water vapour**, so the vapour channel transmits energy to the stone ("parting the water").
- Stone-absorbed photons cause melting, chemical decomposition or **superheated interstitial water → localised microexplosion**.
- **Most Ho:YAG, TFL and pulsed Tm:YAG lasers fragment photothermally**. Pulses of hundreds of μs **preclude stress confinement**; time-resolved imaging and pressure studies show **no significant photomechanical effect**.

LASER PRESSURE TRANSIENTS

Source	Peak transient
Photomechanical lasers	>300 bar
Ho:YAG, pulse energy <1.0 J (usual clinical range)	8–20 bar
Ho:YAG at 2.0 J	20–40 bar
Any laser >250 μs	Typically <20 bar

Further evidence for the photothermal mechanism

- **Moses dynamics on high-speed imaging:** the first ~60 μs vaporise the water between fibre and stone, after which energy couples efficiently through the vapour. **Photomechanical lithotripsy requires water; photothermal lithotripsy loses energy to water.**
- **Incidence angle: perpendicular strike (0°, normal incidence) → most ablation; fibre parallel to the surface (90°) → large bubble, no ablation** (rare exception: under tightly controlled parallel geometry, water jets can impinge on the stone). **Incident angle correlates with ablation volume.**
- **Thermal breakdown products** are detected during Ho:YAG lithotripsy; **room-temperature stones ablate better than cold stones; pulse-modulated Ho:YAG also works photothermally.**
- **Pulse energy correlates roughly linearly with crater volume.**

Figure legends

- **Fig. 19.8** (Chan, Pfefer, Teichman, Welch, *J Endourol* 2001) — ablation mechanism across pulse duration: (A) ~10 μs , **purely photoacoustic:** large-fragment dissociation plus internal fractures; (B) photothermal ablation with internal fractures; (C) ~200 μs , **localised photothermal ablation without collateral damage — the desirable outcome;** (D) ~20 ms, photothermal ablation **with extensive adjacent thermal damage.** Water or liquid **enhances mechanical damage** through cavitation and better acoustic coupling.
- **Fig. 19.9** (Sea, Jonat, Chew et al., *J Urol* 2012) — pressure transients for struvite (magnesium ammonium phosphate hexahydrate, MAPH) stones vs still water; 100-W Ho:YAG, 365- μm fibre. Peak $\approx$ **20 bar at 1 J, $\approx$ 30 bar at 2 J.** Still water gives lower transients than a stone at the near boundary.
- **Fig. 19.10** (Lee, Ryan, Teichman et al., *J Urol* 2003) — Ho:YAG (2.1 μm , 250 μs FWHM) single-pulse craters in plaster-of-Paris phantoms, optical coherence tomography. **Smaller fibre (272 μm): narrow, deep craters; larger fibre (940 μm): wide, shallow craters. Higher pulse energy → larger crater volume.**

IN PLAIN TERMS — THULIUM VERSUS HOLMIUM

- The thulium fibre laser wavelength is **absorbed several times more strongly by water** than holmium, so it dumps its energy in a shorter distance. That favours **fine dust**.
- Thulium delivers its energy with **lower peak power** (a longer, gentler pulse), so **each pulse kicks the stone less:** less retropulsion.
- Holmium gives a harder punch per pulse: more kick, and more fragmenting than dusting.
- Modern high-power holmium machines with pulse modulation have narrowed the gap, which is why the two trials reached different conclusions.

Thulium fibre laser (TFL)

- **Diode-pumped fibre laser at 1.9 μm — more highly absorbed by water than Tm:YAG (2.0 μm) or Ho:YAG (2.1 μm).**
- Pulse range **50 μs to 20 ms;** typical dosimetry **700–800 μs at low peak power — low peak power explains its lower retropulsion.**
- Fibre-laser architecture allows more pulse-parameter flexibility than multimode emitters — a **potential future advantage** in pulse modulation.

THE TWO DOMINANT LITHOTRIPSY LASERS: HO:YAG VS TFL

	Ho:YAG	TFL
Wavelength	2.1 μm	1.9 μm (1940 nm)
Water absorption	High	Higher
Platform	20 W → 120 W over 25 years; variable pulse durations; pulse modulation (two pulses with synchronised delay)	Wide pulse range; flexible modulation potential

	Ho:YAG	TFL
Mechanism	Photothermal	Photothermal
Fragments (preclinical)	Small	Smaller, faster fragmentation
Retropulsion	Greater	Less (preclinical and clinical)

RANDOMISED CONTROLLED TRIALS (RCTS): TFL VS HO:YAG

Trial	Comparator	Endpoints	Verdict
First RCT	Low-power (30 W) Ho:YAG	Stone-free rate (renal), operative time, bleeding	TFL superior: higher stone-free rate, shorter operative time, less intracorporeal bleeding
Second RCT	High-power (120 W) pulse-modulated Ho:YAG	Stone-free rate, subjective retropulsion, radiant energy	Stone-free rates equivalent regardless of location, size or machine; TFL less subjective retropulsion; Ho:YAG used less radiant energy

- The trials disagree because **the comparator differed**: TFL beats *low-power* holmium, not *modern high-power pulse-modulated* holmium.

Single-wavelength principle

- **One wavelength can produce every effect**, depending on pulse duration:
 1. **Short pulse (<10 µs) → cavitation and shock waves.**
 2. **Pulse of hundreds of µs → localised photothermal ablation.**
 3. **Pulse >20 ms → photothermal ablation with collateral thermal damage.**
- **Landmark experiment, Jansen et al., 1996** (Ho:YAG in still water, no stone, high-speed imaging): **500 ns (Q-switched): large spherical bubbles, single-locus collapse, strong pressure ripples**; longer pulses: less spherical, multi-locus collapse, fewer ripples; **260 µs: pear-shaped, asymmetric collapse, no ripples.**
- Some Ho:YAG, TFL and Tm:YAG platforms deliver pulses **as short as 50 µs**, where **both mechanisms occur simultaneously**.
- **The laser-stone interaction mechanism depends on the dosimetry settings, not merely on the platform.**

Safety Issues

IN PLAIN TERMS — HEAT AND RETROPULSION AT THE BEDSIDE

- **Heat:** laser energy that does not enter the stone heats the irrigation fluid. A calyx is a small, nearly closed space, so **high power with poor outflow turns it into a kettle**, hot enough to injure urothelium.
- The remedies all follow from that picture: **lower power, pauses, more inflow, an access sheath for outflow, and the fibre on the stone.**
- **Retropulsion** is the stone being kicked away by each pulse. **The size of each kick is set by the energy per pulse**, not by how many pulses per second.
- This is why **"dusting"** uses low energy at high frequency, and **"fragmenting"** uses high energy at low frequency and accepts more kick.

Heat generation

- High-power platforms put more energy into the collecting system per unit time; because mid-infrared is water-absorbed, **irrigant in the ureter or calyx can reach cytotoxic temperatures.**
- **Strategies to limit dangerous heat: relatively low power settings; frequent duty-cycle breaks; increased flow** (hand irrigation devices, pressure bags, **small-diameter fibres** to free the working channel); **ureteral access sheath** for outflow drainage; **fibre tip kept close to the stone** (photons into the stone mean faster lithotripsy; photons into water mean more total energy is needed).

- **Ho:YAG and TFL can injure urothelium directly** if misdirected into soft tissue.

Retropulsion

- Stones are **free-floating**; displacement forces the surgeon to chase them and adds operative time.

DETERMINANTS OF RETROPULSION

Parameter	Effect on retropulsion
↑ Pulse energy	↑ Retropulsion
↑ Pulse frequency (at low pulse energy)	No increase
↑ Pulse duration / ↓ peak power	↓ Retropulsion

Concept — pulse duration, not wavelength, decides the mechanism. Short pulses confine stress and break stones mechanically with high-pressure bubbles; long pulses cannot confine stress and ablate thermally through a vapour channel. Every clinical difference between lasers — fragment size, retropulsion, the value of fibre angle, the heat risk — follows from which side of that line the pulse falls.

Lasers for Soft Tissue

Uses

- Endoscopic management of **urothelial carcinoma**; **benign prostatic hyperplasia (BPH)**; external treatment of **genital warts and penile carcinoma in situ**.

IN PLAIN TERMS — PENETRATION DEPTH IN SOFT TISSUE

- **Optical penetration depth is how deep light gets before it is absorbed.**
- **Very shallow penetration** (thulium fibre laser) concentrates all the energy in a thin layer, which **vaporises and chars** (turns black).
- **Deeper penetration** (Nd:YAG) spreads gentler heating through a thick layer, which **coagulates deeply** but risks delayed sloughing of tissue.
- **Scatter** is light bouncing around inside tissue like headlights in fog. It spreads the energy sideways and reduces how deep it goes.
- **Fluence** is simply the energy delivered per unit area of tissue.

TISSUE OPTICAL PROPERTIES AND LASER CONFINEMENT

Tissue property	Effect
Strong scatter	Less absorption at depth; penetration attenuates steeply
Weak scatter	Penetration limited by absorption or beam diffraction
High absorption + low scatter	High spatial confinement
Low absorption + high scatter	Large penetrated volume, low fluence
Wide beam (same power)	Lower fluence at a given depth than a narrow beam
Pigmentation	Pigments preferentially absorb certain wavelengths and draw energy to themselves

- The **absorption coefficient** measures chromophore absorption; higher absorption → thermal change → **vaporisation or coagulation**. It varies with wavelength and so shapes the fluence distribution.
- **Optical penetration in water: Ho:YAG 0.31 mm vs TFL 0.08 mm.** Penetration is non-linear: at **1 mm**, **Ho:YAG retains a few percent of its energy; TFL retains essentially none** — true in tissue too, because of its high water content.

Porcine incision studies

- **Zones from centre outward:** vaporisation → carbonisation → irreversible damage → reversible damage.

SOFT-TISSUE INCISION CHARACTERISTICS BY LASER

Comparison	Incision depth	Carbonisation	Coagulation / area
Ho:YAG vs TFL	Ho:YAG deeper at most settings	Ho:YAG none; TFL consistent carbonisation (matches clinical charring with TFL)	Coagulation zones not significantly different
Ho:YAG vs Tm:YAG	Ho:YAG deeper	—	Tm:YAG greater total lased and coagulation areas

- **Commonest endoscopic soft-tissue lasers:** Ho:YAG, Tm:YAG, KTP/lithium triborate (LBO), TFL — used for dissection or ablation.
- Absolute depths depend on the experimental setup, but **trends follow laser physics**.

TABLE 19.2 COMMON LASERS USED FOR SOFT TISSUE IN UROLOGY AND THEIR CHARACTERISTICS

Laser	Wavelength (nm)	Optical penetration (mm)	Output
KTP/LBO	532	0.8	Continuous or quasi-continuous
Nd:YAG	1064	10	Continuous or quasi-continuous
TFL	1940	0.08	Pulsed
Tm:YAG	2000	0.2	Continuous (pulsed versions also exist)
Ho:YAG	2100	0.31	Pulsed
Diode (text)	940–1490	0.5–5	—

- What the table is for: **penetration depth predicts behaviour** — deep (Nd:YAG) means coagulation at depth with a risk of delayed necrosis; very shallow (TFL, Tm:YAG) means precise vaporisation with char; intermediate (KTP/LBO, Ho:YAG) sits between.
- The pattern: **water-absorbed mid-infrared lasers penetrate least**; near-infrared Nd:YAG, absorbed strongly by neither water nor haemoglobin, penetrates most.
- The trap: **penetration is not monotonic with wavelength** — KTP at 532 nm is shallow because of haemoglobin absorption, not because of its wavelength per se.

IN PLAIN TERMS — WHY GREEN LIGHT FOR THE PROSTATE

- An object looks red because it **absorbs green light and reflects red**. Blood and a vascular prostate therefore absorb green light avidly.
- GreenLight energy is deposited exactly where the blood is, **vaporising tissue while sealing vessels**, which is why it can be used in patients who continue anticoagulation.
- **"Side-firing"** means the fibre tip deflects the beam at an angle, so it strikes the prostate wall beside the fibre rather than straight ahead.

KTP/LBO side-firing ("GreenLight") laser

- **Development:** 1990s **60 W → 80 W** (popularised internationally; an **Nd:YAG beam frequency-doubled through a KTP crystal → 532 nm**) → **120 W and 180 W** (LBO crystal instead of KTP, **same 70° side-firing delivery**). All are **pulsed, solid-state lasers**.
- **532 nm (green) is preferentially absorbed by haem**; green's complementary colour is red, so absorption is high in **well-vascularised red tissue such as prostate**.
- **Optical penetration 0.8 mm, but tissue effect 1–3 mm**. Mechanism: **photothermal vaporisation of intracellular water (photovaporisation)**.
- **Haemostasis permits photovaporisation without interrupting anticoagulation**.

- **Dosimetry** (in vitro): vaporisation and coagulation depth **increase with power up to 180 W; 200 W gives no significant further gain**. Best ablation volume per time at **4 mm/s sweep speed** and **≤2 mm working distance**. Clinically, **180 W shortens operating time vs 120 W**.

IN PLAIN TERMS — PHOTODYNAMIC THERAPY

- A **light-sensitive drug** is given; it sits harmlessly in tissue, concentrating in tumour, until light of the right colour switches it on.
- Once activated, it produces a **highly reactive form of oxygen** that damages the cells around it. **No heat and no cutting: the light acts as a trigger, not as a tool.**

Photochemical interaction and photodynamic therapy

- A **third laser mechanism**, after photomechanical and photothermal: **light catalyses a chemical reaction in tissue**.
- **Photodynamic therapy**: photosensitising dye → accumulates in target tissue → activated by a specific wavelength → **singlet oxygen** → reacts with nearby biomolecules.
- Used for **urothelial cancer**: a **large tumour volume can be treated with a single intravesical light application**.

KEY POINTS

- **Pulse duration is the key lithotripsy parameter.**
- **Ultrashort (<500 ns) → plasma. Short (>500 ns, <10 μs) → cavitation and shock waves. Long (>250 μs) → photothermal ablation** (direct photon absorption).
- **Lasers generate heat, with risk of operative injury.**
- **Retropulsion rises with pulse energy and falls with longer pulse duration.**
- **Soft-tissue ablation depends on fluence in tissue and on the wavelength's scatter and absorption in that tissue.**

Concept — in soft tissue the question is not what the laser can do but **where its energy stops**. Absorption and scatter by water and haemoglobin fix the penetration depth, and penetration depth fixes the balance of vaporisation, carbonisation and coagulation.

Electrical Energy

IN PLAIN TERMS — ELECTRICITY FROM FIRST PRINCIPLES

- **Electric current** is a flow of charged particles: electrons in wires, and ions (sodium, potassium, chloride) in tissue and body fluids.
- **Voltage** is the push driving the flow. **Resistance** is how hard it is to push through.
- With a garden hose: voltage is tap pressure, current is water flow, and resistance is how narrow or long the hose is.
- **When current is forced through resistance, energy is released as heat.** That is how a toaster element glows, and it is the whole of electrosurgery: **the patient's tissue is the resistance, so the tissue becomes the toaster element.**
- **Direct current** flows one way (a battery). **Alternating current** reverses direction many times per second: mains supply in India reverses 50 times per second; an electrosurgical generator reverses about 500,000 times per second.
- **Impedance** is simply the name for resistance when the current is alternating.

Basics

- Electrosurgery is ubiquitous and its principles are often learned empirically; **understanding the physics prevents misadventure.**
- **Ohm's law (Equation 2):** $V = I \times R$ (voltage = current $\times$ resistance). **Resistance $\propto$ resistivity $\times$ length, and $\propto$ 1/cross-sectional area.**
- **Water-pipe analogy:** voltage $\approx$ water pressure; current $\approx$ volumetric flow; resistance $\approx$ viscosity, pipe length and inverse cross-section.

DIRECT VS ALTERNATING CURRENT

Direct current (DC)	Alternating current (AC)
Constant, one-direction electron flow	Oscillating, sinusoidal; used for efficient long-distance delivery
Opposition to flow = resistance	Opposition to flow = impedance (Z) : a complex number with amplitude and phase, both frequency-dependent
—	Frequency = number of direction reversals per second (Hz); average voltage is less than peak voltage

*Electrosurgery***IN PLAIN TERMS — WHY SURGICAL CURRENT DOES NOT ELECTROCUTE THE PATIENT**

- At **mains frequency (50 Hz)** each push of current lasts long enough to fire nerves and muscles. That is why a mains shock causes spasm and cardiac arrest.
- At **about 500,000 Hz** the current reverses so fast that nerve and muscle membranes cannot respond. The ions simply jiggle back and forth, and **that jiggling becomes heat.**
- This is what makes it possible to pass a surgical current through a patient safely.
- The **obturator jerk during TURBT** is the exception: the nerve lies just outside the lateral bladder wall and can still be stimulated. Arcing at the loop is thought to generate lower-frequency components that nerves can respond to.

FREQUENCY AND TISSUE EFFECT

Low frequency	High frequency (radiofrequency)
Cell membrane depolarisation $\rightarrow$ muscle contraction	Oscillation too rapid to depolarise $\rightarrow$ little or no contraction ; tissue at the active electrode acts as a resistive impedance $\rightarrow$ ionic excitation $\rightarrow$ heating, desiccation, cellular damage, haemostasis

- **Electrosurgical frequency $\approx$ 500 kHz**, in the radiofrequency range — hence "**radiofrequency generator**".
- **History:** William T. Bovie (PhD in plant physiology). **First human use 1926, with Harvey Cushing**, on a previously inoperable vascular parietal tumour; **patent 1931, sold for \$1**; Bovie died impoverished.

IN PLAIN TERMS — CURRENT DENSITY: STILETTO HEEL VERSUS ELEPHANT FOOT

- A woman in stiletto heels can dent a wooden floor that an elephant's broad foot would not, because her weight is concentrated on a tiny area. Electric current behaves the same way.
- At the **small active tip** the current is crowded into a tiny area, so the heat is intense.
- At the **large dispersive pad** the same current is spread over a big area, so the heat is negligible.
- If the pad **peels partly off**, the contact area shrinks, the pad effectively becomes a second "tip", and the patient is **burned under the pad.**

Monopolar electrosurgery

- **Circuit:** radiofrequency generator → **active electrode** (Bovie tip) → patient's tissue → **dispersive pad (return electrode)** → generator.
- **Current is constant around the circuit; current density decides the effect:** **small tip area** → **high current density** → **local heating** (the **tip stays cooler than the tissue**, which absorbs the heat); **large pad area** → **low current density** → **no heating**.
- **Pad must be fully adherent:** partial adherence → smaller contact area → higher current density → **skin burn**. **Inspect the pad site post-operatively**; many units **detect inadequate pad contact and block activation**.

*Cutting, coagulation and blended settings***IN PLAIN TERMS — CUT, COAGULATE, SPRAY, BLEND**

- **Cut** (continuous current): heats cells so fast that the water inside them flash-boils and the cells burst. The result is a **clean split with little charring**, like a very hot knife.
- **Coagulate** (short bursts at higher voltage, off most of the time): heat builds slowly and spreads during the pauses, so cells **cook and dry out rather than burst**, as egg white turns solid. This **seals vessels** and spreads deeper.
- As the tissue dries it conducts less, so coagulation **naturally limits itself**.
- **Spray (fulguration):** voltage high enough for **sparks to jump the air gap**, like small lightning. It chars a surface without touching it and is useful for a diffusely oozing bed.
- **Blend** is a mixture: cutting with some coagulation.

ELECTROSURGICAL WAVEFORMS: CUT, COAGULATION, SPRAY, BLEND

Setting	Waveform	Voltage / current	Tissue effect
Cut	Continuous sinusoid	Lower voltage, higher current; typically 50–80 W	Rapid heating to boiling → clean vaporisation, little char
Coagulation (contact)	Interrupted ~30,000 times/s; ~6% duty cycle	Higher voltage, lower current	Tissue cools in off-cycles, heat conducts deeper → thermal denaturation below boiling → electrodesiccation → rising impedance self-limits current; spread up to 3 mm
Spray fulguration	Interrupted	Higher still	Voltage ionises the air gap (spark gap) → non-contact thermal delivery
Blend	Cut waveform as base, with interruptions	Intermediate	Cutting with enhanced coagulation

Generator settings

- Most generators let the surgeon choose **wattage** for cut and coagulation, with preset waveforms; on many, "**cut**" is **actually a blend** with adjustable interruptions.
- **Recommended:** cut 50–80 W; coagulation 30–50 W.

IN PLAIN TERMS — WHERE THE CURRENT GOES

- **Monopolar:** the current travels **through the whole patient** from tip to pad, so there are many possible stray paths.
- **Bipolar:** the current travels only **between the two jaws** through the tissue held between them. It is confined and needs less power.

- **Electrocautery** is a **heated wire**, like a soldering iron. The current heats the wire and never enters the patient.

Bipolar electrosurgery and electrocautery

MONOPOLAR VS BIPOLAR ELECTROSURGERY VS ELECTROCAUTERY

	Monopolar electrosurgery	Bipolar electrosurgery	Electrocautery
Electrodes	Small active electrode + large distant dispersive pad	Two electrodes of about equal size , both grasping tissue	Heated metal implement (often a loop)
Patient in circuit?	Yes, whole path to the pad	Yes, only tissue between the jaws	No
Heat arises	Tissue at the tip (current density)	Tissue between both electrodes	The implement itself , which then heats tissue
Power	Higher	Lower → less collateral damage , controlled field	Low ; battery handheld devices, mostly ambulatory clinics
Safety profile	Pad burns, current leaks, capacitive coupling	Can complete the circuit with no tissue if the tines touch	Limited : hot tip, maintenance, battery storage

- **The operating-room "Bovie" is correctly an electrosurgical unit, not an electrocautery.**

Electrosurgical devices

- Mostly **bipolar jaws** combining **electric current with mechanical compression** for haemostasis; a "**tissue response generator**" adjusts output as clamped tissue desiccates and its conductivity changes.
- Current flows **only between the jaws**, with no dispersive pad; open and laparoscopic handpieces exist.
- **LigaSure (Medtronic)**: open and robotic prostatectomy (pedicle sealing) and cystectomy; claimed advantages in operative time and positive margin rate; **cost limits use**.

Safety Issues in Electrosurgery

- **Injury is rare and largely preventable.**

Surgical fire

- Requires **flammable material + electrosurgical ignition**: drapes, preparation solutions, **oxygen-rich anaesthetic gas**. Ignition by **inadvertent activation** or **direct coupling** to a nearby metal instrument.
- **Electrosurgery is implicated in the majority of operating-room fires**, which **most often involve the head and upper torso**, usually with **supplemental oxygen**.

Current leaks and the dispersive pad

- **Insulation failure** lets the circuit contact metal, water, the patient or the surgeon.
- **The dispersive pad is a common injury site** — technically a "**dispersive electrode**". **Avoid** bony prominences, scars, dense hair, implants, **pacemakers**, and breast or orthopaedic implants.
- **Orthopaedic prostheses are not a contraindication**: place the pad **away from the implant**, e.g. **contralateral to a hip prosthesis**. **Remove metal jewellery**; if impossible, never place the pad over it.

Inadvertent activation

- Leaning on the hand-switch or stepping on the foot pedal. **Keep spatial awareness of the devices; holster active electrodes when not in use.**

IN PLAIN TERMS — ELECTRICITY THAT JUMPS THROUGH INSULATION

- Alternating current can **induce a current in a nearby metal object without touching it**, passing through the insulation between them.

- Cords wrapped together, or wrapped around metal instruments, set this up, and **current can appear where no one expects it.**

Capacitive coupling

- A **capacitor = two conductors separated by an insulator**; if the charge difference exceeds the insulator's capacity, current jumps across.
- Examples: **monopolar and bipolar cords wrapped together; cords wrapped around metal instruments.** **Active electrode monitoring systems detect insulation failure and capacitive coupling.**

Surgeon injury and personnel responsibilities

- **Surgical gloves have low electrical resistance**; high voltage penetrates **weakened or damaged gloves. Use the lowest current that achieves the effect.**
- Perioperative personnel: equipment in date and maintained; jewellery removed; **checklist for implants, prostheses and implantable cardiac devices.**

Endoscopic Use of Electrosurgery

- Main uses: **transurethral resection of the prostate (TURP)** and **transurethral resection of bladder tumour (TURBT)**, each available in monopolar and bipolar versions.

IN PLAIN TERMS — WHY GLYCINE, WHY SALINE

- **Monopolar resection** needs the current to leave the loop, pass through the prostate and travel to the pad. If the irrigant were saline (full of sodium and chloride ions, a good conductor), the current would **leak away through the fluid instead of cutting**, like trying to cut under seawater. Hence a **non-conducting fluid such as glycine.**
- The price: glycine absorbed into the circulation **dilutes blood sodium** (the transurethral resection syndrome). This is why resection time is limited, especially in vascular glands.
- **Bipolar resection** deliberately uses saline, because the current only needs to pass from the loop through the fluid to a return electrode on the same instrument.

MONOPOLAR VS BIPOLAR TRANSURETHRAL RESECTION

	Monopolar TURP/TURBT	Bipolar TURP/TURBT
Circuit	Loop → patient → dispersive pad → generator	Completed within the instrument ; one common North American design energises the surrounding saline into a plasma layer , so the patient is not in the circuit (technically closer to electrocautery)
Irrigant	Ion-free : glycine (typically 1.5%, North America and UK), sorbitol-mannitol (continental Europe)	Isotonic saline
Hyponatraemia	Increased (absorptive); limit resection time, especially in vascular glands	Reduced
Efficacy / safety	Reference	Comparable efficacy, favourable safety (moderate-quality evidence, short follow-up)
Obturator reflex	Risk	Thought to be reduced ; bladder perforation data conflicting

	Monopolar TURP/TURBT	Bipolar TURP/TURBT
Typical settings	Cut 180 W; coagulation 80–120 W	—

- **Why monopolar irrigant must be ion-free:** ions disperse current away from the loop and reduce its effect, so normal saline **cannot** be used; the irrigant must also be **relatively inert, non-toxic and transparent**.
- **Sheath insulation:** the loop is **insulated except at its tip**; damaged insulation lets current pass through the metal sheath → **loss of cutting effect or urethral injury**. **Inspect loop insulation before mounting it on the bridge**.

Obturator reflex

- Risk **greatest at the lateral bladder wall**: current stimulates the obturator nerve → **sudden leg adduction** → **bladder perforation, under general or spinal anaesthesia**.
- **Prevention: neuromuscular blockade** — depolarising (succinylcholine) or non-depolarising (rocuronium) — with **timing coordinated with the anaesthetist**, especially with short-acting agents; **selective obturator nerve block**; **avoid overfilling the bladder**; **intermittent activation**; **bipolar resection**.

IN PLAIN TERMS — EMDA

- An electric field across the bladder wall **drives charged drug molecules into the tissue**, rather than waiting for them to soak in.
- It also makes cell membranes temporarily more permeable.

Electromotive drug administration (EMDA)

- Enhances **intravesical mitomycin C (MMC)** penetration into the lamina propria, otherwise poor. Mechanisms: **electroporation, iontophoresis, electro-osmosis/electrophoresis**.
- Set-up: drug instilled → **active electrode in the bladder + dispersive pads on the abdomen** → electric field → **higher tissue drug concentration**.

EMDA: EVIDENCE

Source	Finding
RCT	EMDA-MMC + bacille Calmette-Guérin (BCG) vs BCG alone: improved recurrence-free and progression-free survival
Meta-analysis	Recurrence-free survival improved; progression-free survival not improved ; side-effect profile favourable in some domains
Cochrane review	Evidence low or very low quality; magnitude of benefit uncertain

IN PLAIN TERMS — MICROWAVE, RADIOFREQUENCY, HEAT SINK AND CHEMOHYPERTHERMIA

- **Microwave ablation works like a microwave oven**: the field makes water molecules flip back and forth billions of times per second, and that motion becomes heat inside the tissue.
- **Radiofrequency ablation** passes current from the probe into the surrounding tissue, which heats as a toaster element does. The probe itself does not heat; the tissue does.
- If the tissue **chars**, the charcoal stops conducting, current stops flowing, and ablation is incomplete. That is why 150 °C is undesirable.
- **Heat sink**: flowing blood carries heat away, as coolant carries heat from a car engine. A tumour next to a large vessel is harder to "cook". Microwave energy heats faster and wider and so is less defeated by this.

- **Chemohyperthermia:** gently warming the bladder wall to about 42 °C makes cell membranes leakier, so more mitomycin enters. Cancer cells also tolerate heat worse than normal cells.

Microwave thermotherapy (soft tissue)

- Microwave ablation (MWA). Mechanism: **dielectric hysteresis** — molecules repeatedly realign with the oscillating electric field → rotational and vibrational kinetic energy → **heat**.
- Frequencies **915 MHz or 2.45 GHz**: generator → coaxial line → **antenna(e) placed directly in the tumour**. **The shaft heats**, so a **cooling system** protects the tract.
- Urology: **alternative to radiofrequency ablation (RFA) for renal tumours**. **Heat sink**: the highly perfused kidney carries heat away; **MWA reaches larger zones and higher temperatures than RFA and is less susceptible to heat sink**.
- Early animal concerns — **collecting-system absorption** and **inconsistent zones** — but **safety and primary efficacy are comparable to RFA**.

Radiofrequency in urology

- **AC at 375–900 kHz**; the probe sits at the **centre of the lesion**. **The probe does not generate heat**; **Joule heating arises in the tissue** from ionic excitation and conducts outward **spherically**.
- **Target 60 °C** for reliable cell death; at **150 °C**, **carbonisation** creates **high impedance and non-uniform heating**. **Temperature probes** monitor the time–temperature plan.
- **Small renal masses**: RFA is **less common than cryoablation in the United States**, more common in Germany; **oncological outcomes are equivalent to cryoablation**.
- **Chemohyperthermia for non-muscle-invasive bladder cancer (NMIBC)** — device-assisted intravesical therapy like EMDA. Intravesical mitomycin is given **while the bladder wall is heated to 42 °C**: **increased cell permeability → more drug absorption**; **cancer cells are more heat-sensitive**; **heat-shock protein release may potentiate mitomycin**.
- **Synergo** (Medical Enterprises Group, Amsterdam) is the commonest system: a **Foley-type catheter with a 915-MHz emitter and thermocouples**.

DEVICE-ASSISTED INTRAVESICAL THERAPY: EMDA VS CHEMOHYPERTHERMIA

	EMDA	Chemohyperthermia
Energy	Electric field (electrode in bladder, pads on abdomen)	Microwave-band energy (915 MHz) via catheter
Mechanism	Electroporation, iontophoresis, electro-osmosis	42 °C heating → permeability, heat sensitivity, heat-shock proteins
Drug	Mitomycin C	Mitomycin C
Indication	NMIBC	NMIBC

RENAL-MASS THERMAL ABLATION: RFA VS MWA VS CRYOABLATION

	RFA	MWA	Cryoablation
Energy	AC 375–900 kHz; Joule heating in tissue	915 MHz / 2.45 GHz; dielectric hysteresis	Joule–Thomson gas expansion (argon)
Target	60 °C ; avoid 150 °C char	Higher temperatures, larger zones	<–40 °C, two freeze–thaw cycles
Heat sink	Susceptible	Less susceptible	—
Monitoring	Temperature probes	—	Visible ice ball on imaging
Outcomes	Equivalent to cryoablation	Comparable to RFA	Commonest in the United States

KEY POINTS

- **Monopolar circuit:** generator → active electrode → patient → dispersive pad → generator.
- **Unintentional current leaks injure patient, surgeon or staff; the dispersive pad is a common site.**
- **Monopolar TURP/TURBT irrigant must be ion-free.**
- **Ion-free irrigants carry increased hyponatraemia risk.**
- **Lateral-wall TURBT risks obturator stimulation → adduction → bladder perforation.**

Concept — current density is the whole of electrosurgery: the same current burns where the contact area is small (the tip) and is harmless where it is large (a well-applied pad), and every pad burn, bipolar advantage and insulation failure is a failure or exploitation of that ratio. The **waveform, not the power, then chooses between cutting and coagulation:** continuous heating vaporises, interrupted heating denatures.

ELASTIC WAVES**IN PLAIN TERMS — SOUND VERSUS LIGHT FOR TREATMENT**

- **Light is absorbed within millimetres** in water and tissue, so a laser must be taken to the target on a fibre.
- **Sound travels centimetres through tissue** with little loss, so it can be **sent in through intact skin and focused deep inside:** shock wave lithotripsy and HIFU need no incision.

ACOUSTIC VS LASER ENERGY

Acoustic (elastic)	Laser (EM)
Greater radiation force for the same energy	Lower radiation force
Lower intensity / energy density at the source	Higher intensity
Deeper tissue penetration	Short absorption depth (water, tissue, stone)
Needs a large source at the skin to focus energy at depth	Delivered by fibre to the target

Applications

- **Lithotripsy of calcifications:** shock wave lithotripsy (SWL); shock wave therapy (SWT); burst wave lithotripsy (BWL, **investigational**); ultrasonic propulsion (**investigational**); intracorporeal acoustic and pneumatic lithotripters.
- **Tissue ablation:** thermal high-intensity focused ultrasound (HIFU); mechanical histotripsy (**investigational**).
- **Therapeutic vs diagnostic acoustics:** therapy uses **longer durations (for heating)** and/or **greater positive or negative pressures** (to crack brittle material or cavitate fluid); source usually at the skin, target centimetres deep, sometimes via a **catheter-like probe**.

Advantages of acoustics and coupling

- **(1) The same energy both images and treats. (2) Transcutaneous treatment** of a millimetre- to centimetre-sized region deep in the body.
- The source is **coupled by water or gel**, because **air-tissue impedance mismatch** prevents transmission. Intracorporeally, the source is miniaturised or the wave runs down a **waveguide**, coupled by urine or direct stone contact.

IN PLAIN TERMS — HOW A LITHOTRIPTER MAKES ITS BANG

- A **transducer** converts one form of energy into another. A loudspeaker turns electricity into sound; a microphone turns sound into electricity; an ultrasound probe does both.
- **Electrohydraulic**: a spark in water, like an underwater firecracker.
- **Electromagnetic**: a coil pushes a metal membrane, like a very powerful loudspeaker cone.
- **Pneumatic**: one part strikes another, a miniature jackhammer.
- **Piezoelectric**: special crystals that **change thickness when a voltage is applied** (and make a voltage when squeezed, the spark in a gas lighter). Many crystals together push the water.
- **Focusing** with a curved reflector or lens works like a satellite dish gathering signal to one point.

Transducers

- A **transducer** converts energy from one form to another (here electrical → mechanical).

ACOUSTIC TRANSDUCER TYPES

Type	Mechanism	Focusing	Signal
Electrohydraulic	Underwater spark vaporises water → spherical wave; abrupt expansion creates a shock at the source	Ellipsoidal reflector	Impulse
Electromagnetic	Impulse through a coil pushes a thin plate (membrane) against the coupling fluid	Lens or parabolic reflector ; shock forms during propagation to the focus	Broadband impulse
Pneumatic	One part impacts another	Waveguide (rigid tapered catheter) delivers energy directly into the stone	Impulse
Piezoelectric	Poled ceramic ; compression → voltage; applied oscillating voltage → thickness vibration	Bowl shape or lens (focus at centre of curvature), or waveguide	Only type producing both long narrowband bursts and impulses

Piezoceramic details

- **Most efficient at the frequency whose half-wavelength fits the element's thickness**; a **matching network** matches transducer and source impedance.
- **Primary limitation: lower frequency → thicker element** → higher voltage and larger lateral dimensions → **larger source**; if thickness approaches width or length, **parasitic vibration modes** reduce efficiency; **excess voltage cracks the element**.
- **Waveguide** works like a **reverse trumpet**: it concentrates energy from horn to mouthpiece but **filters broadband input to a few primary tones**.
- **Typical focus: <1 cm transverse, a few cm axial**. Treating a larger region at millimetre-scale distance with one beam requires **frequencies <1 MHz**.
- **Phased arrays**: multi-element ceramics allow **electronic beam steering** without moving the source, but are **more complex and expensive**; most sources are single-element.

Figure legend

- **Fig. 19.14** — Shock wave lithotripter designs: (A) **electrohydraulic** — spark at one focus of an ellipsoidal reflector; the reflected half converges on the **second focus, placed on the stone by fluoroscopy**; (B) **electromagnetic** — cylindrical source radiating laterally into a **parabolic reflector** (other designs use a spherical source or lens); (C) **piezoelectric** — roughly spherical array focused at the **centre of curvature** on the stone.

IN PLAIN TERMS — HEAT VERSUS BUBBLES

- **Higher frequency:** the tissue absorbs more of the wave, so there is **more heating** (and shallower reach).
- **Lower frequency:** each pulling (negative-pressure) phase lasts longer, giving bubbles **more time to grow**, so there is **more cavitation**.
- This single asymmetry explains why heating devices and bubble-making devices choose different frequencies.
- **"Threshold effect":** cavitation does not build up gradually. Below a certain negative pressure nothing happens; above it, bubbles form.
- The **twinkling artefact** you see behind stones on Doppler comes from tiny bubbles on the stone surface.

Acoustic interaction with tissue

- Mechanisms: **heating, cavitation (bubble activity), mechanical stress, radiation force, and possibly undetermined non-thermal processes.**
- **Heating:** absorption along the path → local heat; **focusing raises amplitude, so bioeffects concentrate at the focus by design.** Example: **transrectal focusing into the prostate necrotises tumour while sparing the rectal wall;** the focus can be moved to successive targets and necrosed tissue is **resorbed**. Heat can also **seal vessels (ischaemic kill)** or permit **bloodless resection**. **Propagation and absorption are calculable**, so injury is predictable and targetable.
- **Frequency trade-off:** too high a frequency → absorbed before the focus. Nonlinear shocks forming only near the focus carry high-frequency components → **dramatically increased local heating.**
- **Boiling histotripsy:** shock-enhanced heating → **~100-μm focus boils within milliseconds** → millimetre vapour bubble → field shakes surrounding tissue → **mechanical shredding to submicron particles** → tissue flows into the bubble and **atomises**. **Mechanical liquefaction, not thermal necrosis.**
- **Cavitation and gas-body activation** are driven by the **negative pressure amplitude: cavitation lowers pressure; boiling raises temperature.** Therapeutic negative pressures (**MPa order**) exceed **10× atmospheric (0.1 MPa)**; the **tensile stress** tears the medium into a rapidly expanding bubble.
- **Nucleation sites:** interfaces, particles (stone surfaces, debris), pre-existing gas. **Microbubbles on stones cause the twinkling artefact; lung and bowel gas** make these organs vulnerable. **No SWL within 24 hours of microbubble ultrasound contrast.**
- **Cavitation injury:** the bubble or cloud **grows during one negative phase and stretches and tears tissue;** **cloud-cavitation histotripsy** liquefies tissue. **Near solid surfaces, collapse re-radiates shock waves and fluid jets through the collapsing bubble** strike and crack the surface.

HEATING VS CAVITATION — THE GOVERNING ASYMMETRY

Heating	Cavitation
Increases with frequency (absorption)	Increases as frequency decreases (longer negative phase, longer tension)
Dose-dependent	Threshold effect

- **Secondary effects: vasoconstriction, ischaemia, extravasation, reperfusion injury** — may exceed the direct insult, harmful or beneficial. **Sub-injurious shock waves cause protective renal vasoconstriction, but too many shocks turn vasoconstriction and reperfusion into injury.**

IN PLAIN TERMS — WHY STONES CRACK AND KIDNEYS SURVIVE

- **Stones are brittle**, like chalk or glass: they cannot stretch, so a strong pull cracks them.

- **Tissue is elastic**, like rubber: it stretches and springs back. Bend a stick of chalk and it snaps; bend a rubber tube and it recovers.
- **How the stone breaks**: the shock travels faster through the stone than through the fluid around it. Waves converge at the **far side** of the stone and pull it apart there, so **the far end cracks off first**. Each further shock extends the cracks, as bending a paper clip repeatedly finally snaps it (fatigue).
- The beam must be **wider than the stone**, because part of the effect comes from the wave running along the stone's outer surface.

Physics of acoustic lithotripsy

- Focused shocks fracture the stone; repeated shocks progressively comminute it into passable fragments — **thousands of shocks over ~45–60 min.**
- **Primary advantage: non-invasive transcutaneous delivery**; in the United States usually performed in a surgical suite under sedation or general anaesthesia.
- **Lowest morbidity and complication rate** of the three lithotripsy techniques (SWL, ureteroscopy, percutaneous nephrolithotomy), **but SWL causes renal trauma**, varying with device, kidney size and protocol. **Unpredictable effect; incomplete breakage → repeat procedures.**
- **Poor fragmentation predictors: hard compositions (calcium oxalate monohydrate, cystine, brushite); Hounsfield density >1000.**
- **Fracture mechanism** (beam must be **larger than the stone**): shock enters the stone (faster, higher sound speed) → fluid wave along the surface generates **shear waves**, reinforced by the **Mach wave** → angled shear waves **converge on the axis at the distal end** → incident wave **reflects and inverts at the distal face (spallation)** → **all tensile stresses meet** → **maximal tensile stress distally** → **first fracture where focused shear and reflected longitudinal waves meet** → cracks grow over successive shocks → **distal end cleaves off.**
- **Cavitation** creates surface pits into which internal cracks grow by **dynamic fatigue**; surface stress is smaller and localised but **grinds fragments smaller**. Once fragments approach the **pulse length**, internal stresses no longer focus and **surface cavitation continues grinding. Too much cavitation shields the stone.**
- **Rate and shielding: faster delivery → larger cavitation cloud** (bubbles lack time to dissolve; each growth-collapse cycle transfers gas inward, increasing the **void fraction**) → the cloud **shields the stone**. Hence **2 Hz is less effective and more injurious than 1 Hz or 0.5 Hz.**

FIG. 19.15 HOW SWL BREAKS STONES (SAPOZHNIKOV, MAXWELL, MACCONAGHY, BAILEY, J ACOUST SOC AM 2007)

Time	Event
0 μ s	Positive pressure spike approaches the stone
1 μ s	Wave inside the stone outruns the fluid wave (higher sound speed)
2 μ s	Fluid wave along the surface → shear waves (tension , shown positive by convention)
3 μ s	Shear waves converge distally; incident wave races ahead
4 μ s	Reflection and inversion at the distal face → spallation wave travelling back
5 μ s	All tensile stresses meet → maximum tensile stress → crack initiation

- **Most solids fail in tension. A beam narrower than the stone produces no surface wave, so a beam broader than the stone is required.**

Specific Acoustic Devices and Applications

- **Hard targets:** SWL, SWT, BWL, intracorporeal lithotripsy, ultrasonic propulsion. **Soft targets:** calcifications and soft tissue.
- **Lithotripsy:** extracorporeal SWL and BWL (investigational) and intracorporeal acoustic/pneumatic devices fragment stones; **ultrasonic propulsion repositions** stones transcutaneously.

IN PLAIN TERMS — WHY SLOWER SHOCK WAVE LITHOTRIPSY WORKS BETTER

- Each shock leaves behind a **cloud of tiny bubbles that needs about a second to dissolve**.
- Fire too fast and the next shock hits that bubble cloud, which **absorbs and scatters it as fog blocks headlights**. Less energy reaches the stone (poorer fragmentation), and the bubbles injure the kidney.
- Hence 60 shocks per minute or slower beats 120 per minute.
- **Rectified diffusion:** a bubble takes in more gas while it is large than it loses while it is small, so it **grows a little with every cycle**.

Shock wave lithotripsy

- A primary stone modality for over 40 years. Machine = **shock source + table + fluoroscopy**.
- **Coupling:** water bath (original) → shallow bath in the table → water-filled balloon with gel on skin (most current). Air bubbles at the coupling interface reduce efficiency disproportionately: a small fractional obstruction at the aperture **substantially reduces peak focal pressure**.
- **Pulse:** peak positive 30–120 MPa (300–1200 bar) with a steep shock front; **positive phase ~1 μs; negative phase ~5 μs; peak negative ~10 MPa; single cycle, broadband; centre frequency ~300 kHz** (below diagnostic ultrasound); pulse energy ~3 J; focal width millimetres to centimetres, length centimetres; **focal intensity >10,000 W/cm²**.
- ~2000 shocks at 0.5–3 Hz (30–180/min).
- **Why stones break but tissue survives:** stones are **brittle** and crack under **tension and shear** (as brick fails in an earthquake); **tissue is ductile**, like building steel; **energy is concentrated at the focus, where the stone lies**.
- **Cavitation bubble life cycle:** high-amplitude, long negative phase **tears water** (and nucleates at tissue interfaces) → **bubble grows by inertia long after the 5-μs pulse, to ~1 mm** → water rushes in → **collapse smaller than original size** → **energy concentrates** → gas/vapour halts collapse → **rebound shock wave** → near a stone, **asymmetric collapse drives a micro-jet into the stone** → **several rebound cycles** → bubble ends **larger than it began (rectified diffusion: more gas enters while large and at low pressure than leaves while small and compressed)**.
- A 1-μm bubble grows to ~30 μm after one pulse and needs ~1 s to dissolve. A next pulse before dissolution re-excites bubbles → **proliferating cavitation cluster** → **tissue injury and shielding**. Slower rate → bubbles dissolve → each pulse reaches the stone.
- **Soft-tissue shock wave therapy (SWT):** soft, fibrous or calcified tissue, e.g. **Peyronie disease**; presumed **cavitation or strain** → **minor injury** → **more pliable repair**. Pneumatic and ballistic sources are marketed for superficial soft tissue on the rationale of **inflammation-driven healing** (analogous to laser dermatology); **less well studied**. Shock waves can also **liquefy tissue like histotripsy**, intentionally or otherwise.

IN PLAIN TERMS — BURST WAVE LITHOTRIPSY

- Instead of **one huge bang**, burst wave lithotripsy sends a **short hum of many gentler oscillations**. The stone vibrates, and the echoes inside it add up to break it.

- Because the pressure is lower, far fewer bubbles form. It can therefore fire faster, and it is **gentle enough to use without anaesthesia**.
- **Fragment size is set by the wavelength**: a higher-pitched probe produces smaller fragments.

Burst wave lithotripsy

- A **small, diagnostic-ultrasound-sized system** for clinic or emergency department use without anaesthesia; handheld or support-assisted probe; ultrasound targeting and real-time monitoring.
- **Piezoceramic, narrowband, 5–100 cycles; peak pressure 5–8 MPa**; energy per pulse about equal to SWL because the burst is longer; **200–1000 kHz**, probes chosen for depth and frequency; beam widths similar to SWL.
- **Less cavitation → higher rate (≥ 10 Hz) → more energy over shorter time**: bubbles stay small, take in less gas and **dissolve in $\sim 1/16$ s**, so the rate can be **$\geq 16\times$ SWL without shielding**.
- **Operators watch for cavitation** (a dynamic white cloud on ultrasound) and **pause if seen**. Lower pressures plus cloud avoidance plausibly explain tolerability without sedation.
- **Coupling by gel is essential**: loss of coupling = loss of image = loss of feedback and efficiency.
- Mechanism: **beam wider than the stone** (≈ 1 cm at the stone); energy enters the lateral and proximal surfaces; **multiple internal reflections of many cycles superimpose → $>5\times$ amplification of the applied negative pressure**; tensile stress similar to SWL at lower applied pressure, at **many points over a longer time**.
- **Fragments are uniform and scale with wavelength (inversely with frequency)**; higher-frequency BWL grinds residual fragments to submillimetre size.

Figure legend

- **Fig. 19.16** — (A) **SWL**: single sharp shock (~ 40 MPa) then a negative trough. (B) **BWL**: multi-cycle sinusoidal burst at lower amplitude. (C) **Large SWL cavitation cloud** vs (D) **few barely visible, short-lived BWL bubbles**. (E) Calculated bubble radius over time (Church model): **SWL bubbles grow to ~ 600 μm and rebound repeatedly**; **BWL bubbles stay ~ 100 μm and die quickly**.

IN PLAIN TERMS — PUSHING STONES WITH SOUND

- Sound exerts a steady push, like wind on a sail.
- Low-pressure, long pulses aimed at a fragment **nudge it along**, for example out of a lower-pole calyx towards the ureter. They break nothing and make almost no bubbles.

Ultrasonic propulsion

- **Same instrument and transducers as BWL, used to reposition rather than break**. Peak positive ≤ 2.4 MPa (24 bar); **~ 1500 cycles** (vs ≤ 100 in BWL) — lower pressure, longer duration.
- **Bursts ≤ 3 s at 50% duty factor**, imaging in the off-time; pulse average intensity **200 W/cm²**; average power over 3 s **~ 10 W**.
- Mechanism: **radiation force** — wave momentum transfers to the stone on reflection; **force = power $\div$ sound speed**; directed **away from the transducer**, so the probe must be manipulated to steer the stone.
- Uses: moving stones during surgery; facilitating passage of **ureteral stones and residual renal fragments**. An RCT: propulsion of residual fragments **substantially reduced relapse** (stone growth, urgent visit or surgery) at 4 years.
- **Little cavitation risk**. Probe and skin warm with long bursts, but it has been **tolerated in all subjects to date**.

ACOUSTIC LITHOTRIPSY DEVICES: SWL VS BWL VS ULTRASONIC PROPULSION

	SWL	BWL	Ultrasonic propulsion
Status	Established (>40 years)	Investigational	Investigational
Waveform	Single-cycle broadband shock	Narrowband burst, 5–100 cycles	Narrowband, ~1500 cycles
Peak pressure	30–120 MPa positive; ~10 MPa negative	5–8 MPa	≤2.4 MPa
Frequency	Centre ~300 kHz	200–1000 kHz	Same transducer as BWL
Rate	0.5–3 Hz (best ≤1 Hz)	≥10 Hz	≤3-s bursts, 50% duty
Cavitation	Large cloud → shielding, injury	Minimal; watch and pause	Negligible
Guidance	Fluoroscopy (sometimes ultrasound)	Real-time ultrasound	Real-time ultrasound
Anaesthesia	Sedation or general (United States)	None	None
Fragments	Variable; end near pulse length	Uniform, $\propto$ wavelength	— (repositions)
Action	Break	Break	Move (radiation force)

Intracorporeal Lithotripters and Contact Sources

IN PLAIN TERMS — CONTACT LITHOTRIPTERS

- Here the energy source actually **touches the stone** through a rigid probe, like a **chisel or jackhammer**. The damage is greatest at the point of contact.
- The same tip against the ureteric or pelvic wall can tear it.
- **Percutaneous nephrolithotomy**: contact probes send elastic waves (impulses or narrowband) down a **stiff waveguide**; **tip displacement produces a jackhammer effect**; stress greatest at the contact point, diverging from there; **spallation** possible; **direct tip contact can tear tissue**. Some ultrasonic devices are thought to break stones into **large fragments by cavitation**.
- **Ureteroscopy**: ultrasonic devices less common; **electrohydraulic (spark) intracorporeal lithotripters persist** — direct wave plus the **water hammer of spark cavitation** → fragmentation of the **proximal stone face**; **bubble expansion and collapse jets can strain and shear the adjacent ureter**.
- **Whole-body vibration** (investigational): skin-contact massagers send diverging waves inward, **mechanically loosening renal fragments for passage**; impulsive skin sources are also used for Peyronie disease.

Soft Tissue Ablation

- **Goal inverted relative to stones**: **ablate tissue**, rather than remove a calculus with minimal tissue injury. Research uses include drug delivery, **renal denervation for hypertension** and tissue softening; **clinical use is mainly ablation**.
- **Two forms**: **thermal (HIFU)** and **mechanical (histotripsy)**.

IN PLAIN TERMS — HIFU COOKS, HISTOTRIPSY LIQUEFIES

- **HIFU is the magnifying glass with sound**. Heat builds only where the beam converges, cooking a **rice-grain of tissue at a time**, and the target is "painted" grain by grain.
- Because cooked and healthy tissue look alike on ultrasound, gaps can be missed.
- **Histotripsy** uses stronger, shorter pulses to create bubble clouds that act as a **microscopic blender**, turning tissue into liquid slurry without cooking it.
- The bubbles are bright on ultrasound, so the **treated zone is visible as it forms**.

High-intensity focused ultrasound

- Diagnostic power has negligible heating; **high intensity → progressive absorption → thermal necrosis.**
- **Frequency similar to imaging of the same target;** focused by a bowl-shaped ceramic or lens; **narrowband pulses of ~1 s.** Lesion ≈ 1 wavelength wide $\times 10$ wavelengths long $\approx$ a grain of rice. Intensity 1500–10,000 W/cm².
- **Higher frequency = more heating at the focus, but more path loss.** Nonlinear shocks → heating within milliseconds.
- **Risks and limits:** necrosis along the beam path; bone or gas in the path increases local heating; necrosed and viable tissue are not distinguishable on standard ultrasound; many exposures are needed for a sizeable region → gaps; alternative strategy: cauterise the feeding vessels.
- **Prostate temperatures exceed 100 °C within seconds,** depending on the absorption coefficient, duration and magnitude.
- **HIFU also acts mechanically:** negative pressure → interstitial bubbles → collapse → cellular damage **compounding the thermal effect.**
- Advantages: **extracorporeal; small focal volume;** co-located imaging and treatment probes allow real-time monitoring. **Approved for BPH and prostate cancer** (transrectal, short focal distance); intra-operative renal devices have existed.

Histotripsy

- **Short but still narrowband pulses at higher intensity than HIFU (>10,000 W/cm²).** Aim: cavitation within one or very few bursts → mechanical liquefaction, not thermal necrosis. First observed with lithotripters.
- **Result: mechanical shredding to submicron fragments;** subsequent pulses shear tissue around the bubbles and even atomise it.
- Being developed for **transcutaneous prostate and kidney cancer;** prostate challenges: **acoustic window** and **fibrous tissue resisting liquefaction;** kidney trials under way.
- **Advantage: the bubble zone is echogenic, visible in real time.** Risks: **off-target bubbles, oversized bubbles → shielding and cavitation injury,** detectable on imaging.

ACOUSTIC SOFT-TISSUE ABLATION: HIFU VS CAVITATION VS BOILING HISTOTRIPSY

	Thermal HIFU	Cavitation histotripsy	Boiling histotripsy
Mechanism	Absorption → thermal necrosis (plus some cavitation)	Large negative pressure tears tissue into a bubble cloud	Shock absorption → boiling within one burst (<100 ms)
Intensity	1500–10,000 W/cm ²	>10,000 W/cm²	>10,000 W/cm ²
Exposure	~1 s narrowband pulses	One or very few short bursts	One burst to boil
Probe	Bowl or lens	Tightly focused → larger probes	Less focused, longer focal distance → smaller probes
Lesion	Coagulative necrosis, rice-grain unit	Submicron liquefaction	Submicron liquefaction, atomisation
Ultrasound visibility	Poor	Echogenic, real time	Echogenic
Status	Approved: BPH, prostate cancer	Investigational	Investigational

Image guidance and operator safety

- **SWL is usually fluoroscopy-guided**, sometimes ultrasound-guided; **most other urological ultrasound therapies are ultrasound-guided**, often from the **same handheld probe**; **magnetic resonance imaging (MRI) monitoring is common elsewhere**. **All probes need liquid or gel coupling**.
- **Ultrasound is non-ionising**: the **only operator risk is at the focus** (transcutaneous) **or the tip of ballistic sources**.

KEY POINTS

- **Gel-coupled skin sources focus energy for a localised internal bioeffect.**
- **Acoustic devices do not transmit through air; they require gel or fluid.**
- **Too high a pulse rate → cavitation cloud → blocked energy delivery.**
- **Acoustic therapies break stones or ablate tissue. Ablation is thermal (absorption) or mechanical (cavitation).**
- **Cavitation amplifies tissue strain and concentrates energy. Cavitation is a threshold effect:** bioeffects occur above a threshold pressure for a given frequency, not below.
- **Fragmentation is primarily crack growth from wave-induced stress.**
- **Acoustic waves are non-ionising; repeated safe exposures do not accumulate into harm. Correctly operated, properly working devices are safe for operators.**

Concept — acoustic therapy is the management of the negative pressure phase. Tension cracks brittle stones and spares ductile tissue, but the same tension nucleates bubbles that either destroy tissue deliberately (histotripsy) or shield the stone and injure the kidney (SWL at high rates). Frequency is the lever: **raise it for heat, lower it for cavitation.**

Direct Contact Energies**IN PLAIN TERMS — STEAM AND WATER JETS**

- **Water vapour thermal therapy (Rezum):** steam scalds more severely than boiling water at the same temperature, because steam releases a large amount of extra heat (latent heat) as it condenses back into water.
- Injected steam therefore **delivers heat quickly and evenly** into prostate tissue without charring, and it stays within the zone it was injected into.
- **Aquablation** is a robot-guided **pressure washer**: a high-speed saline jet cuts tissue mechanically, with no heat.

Fluid-based ablation modalities

- **Water vapour thermal therapy (WVTT) — Rezum (Boston Scientific), for BPH.** Radiofrequency energy **stores thermal energy in water by converting it to vapour**, injected transurethrally into the **transition zone**; it **releases latent heat on condensing** and spreads rapidly and evenly by **convection**.
- Hardware: generator + transurethral device; **18-gauge needle with regularly spaced holes**; delivery pressure **slightly above interstitial pressure** for uniform delivery.
- Vapour **103 °C**; tissue **≥70 °C**; 9-s injection → **spherical zone of cell death**. **No significant thermal gradient**; wet thermal therapy → **no char or carbonisation**.
- Vapour follows the **lowest-pressure path** → **stays within the zonal compartment**, without extension into the peripheral zone on MRI. **Multiple regularly spaced injections** treat the obstructing transition zone in one session; **MRI confirms prostate volume reduction**.

- **Aquablation — AquaBeam Robotic System (Procept BioRobotics):** mechanical, non-thermal ablation by a high-velocity saline jet. Canine histology: a thin layer of swollen cells, then essentially normal tissue. Components: **conformal planning unit, handpiece, 24-Fr cystoscope, biplanar transrectal ultrasound, console.** Sequence: ultrasound mapping of the treatment area → robotic jet delivery.

IN PLAIN TERMS — HOW A CRYOPROBE GETS COLD AND WHY FREEZING KILLS

- **Joule–Thomson effect:** gas escaping from high pressure into low pressure cools sharply, the reason an aerosol can grows cold as you spray it. High-pressure argon expanding inside the probe tip takes it below -100°C .
- **Freezing kills cells in two ways.** Ice crystals form and **pierce membranes**; on thawing, water rushes back into the damaged cells and they **swell and burst**.
- **Two freeze–thaw cycles** repeat both insults, which is why two are used.

Cryotherapy

- For **small renal masses** (principally percutaneous), also applied to **focal prostate cancer therapy**.
- **Freeze–thaw injury (slow freezing):** extracellular ice → extracellular hypertonicity → **osmotic cellular dehydration** → intracellular ice → **membrane disruption** → thaw: extracellular ice melts first → water rushes in → **swelling and bursting**. Rapid freezing skips dehydration; other steps appear analogous.
- **Delayed death:** apoptosis; **microenvironmental ischaemia and inflammatory cytokine release**.
- **Probe cooling: the Joule–Thomson principle** — high-pressure argon expands in the probe → below -100°C → continuous heat extraction.
- **Visible ice ball;** ultrasound monitoring allows adjustment of time and temperature. **Target $<-40^{\circ}\text{C}$; two freeze–thaw cycles with passive thaw.**

PROSTATE ABLATIVE ENERGIES ACROSS THE CHAPTER

Modality	Energy	Mechanism	Distinguishing feature
KTP/LBO photovaporisation	532-nm laser	Photothermal vaporisation	Haemoglobin absorption; continue anticoagulation
Monopolar / bipolar TURP	Radiofrequency electrosurgery	Resection with loop	Irrigant chemistry decides hyponatraemia risk
WVTT (Rezum)	Radiofrequency-generated vapour	Convective heat, latent heat of condensation	No char; zone-confined
Aquablation	High-pressure saline	Mechanical, non-thermal	Robotic, ultrasound-mapped
HIFU	Focused ultrasound	Thermal (plus cavitation)	Transrectal; approved for BPH and cancer
Histotripsy	Focused ultrasound	Mechanical liquefaction	Investigational; fibrous tissue resists

Concept — direct-contact modalities abandon the wave and **carry the energy in the medium itself:** vapour carries heat by convection and condensation, a saline jet carries momentum, a cryoprobe extracts heat. **The tissue boundary rather than the absorption depth limits the lesion**, which is why vapour stays within its zone and the jet leaves only a thin rim of injury.

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